

Leukocyte telomere length and its relation to food and nutrient intake in an elderly population.

BACKGROUND/OBJECTIVES: Shorter leukocyte telomere length (LTL) is associated with several chronic diseases, but only a few studies have assessed the association between dietary factors and LTL. Our objective was to study the association between fats, fruits, vegetables and LTL in a cross-sectional study design. We hypothesized that intakes of fruits and vegetables would be positively associated with LTL and that intakes of fats, and especially saturated fatty acids (SFAs), would be negatively associated with LTL.

SUBJECTS/METHODS: LTL was measured by quantitative real-time polymerase chain reaction in 1942 men and women aged 57-70 years from the Helsinki Birth Cohort Study. We assessed the whole diet by a validated semiquantitative 128-item food-frequency questionnaire.

RESULTS: In general, there were only a few significant results. However, total fat and SFA intake ($P=0.04$ and 0.01 , respectively) were inversely associated with LTL in men adjusting for age and energy intake. In women, vegetable intake was positively associated with LTL ($P=0.05$). Men consuming the most butter and least fruits had significantly shorter telomeres than those consuming the lowest amounts of butter and highest amounts of fruits ($P=0.05$). We found no association between LTL and body mass index, waist-hip ratio, smoking, physical activity or educational attainment.

CONCLUSIONS: In this cross-sectional study of elderly men and women, there were only a few statistically significant effects of diet, but in general they support the hypothesis that fat and vegetable intakes were associated with LTL.